Hyunjae Suh

Ph.D. Candidate in Software Engineering
University of California, Irvine

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RESEARCH INTERESTS

AI for Software Engineering · Large Language Models · LLM Agents for Software Engineering · AI-generated Code Attribution & Provenance · LLM Evaluation · Empirical Software Engineering

RESEARCH EXPERIENCE

Ph.D. Researcher – AI & Software Engineering Sep. 2023 – Present
University of California, Irvine *Irvine, CA*

- Conduct research on applying and evaluating large language models for software engineering tasks, focusing on AI-generated code attribution, accessible code generation, and reliable AI-assisted development.
- Investigated automatic attribution of AI-generated source code using semantic code embeddings, fine-tuned language models, and feature-based classifiers, demonstrating the effectiveness of semantic program representations and achieving an 82.55 F1-score. Published at **ICSE 2025**.
- Conducted an empirical study of prompting strategies (Zero-shot, Few-shot, and Self-Criticism) for accessibility-aware code generation, and proposed *FeedA11y*, a feedback-driven ReAct-based LLM agent that iteratively improves generated code using automated accessibility evaluation. Published at **ACM TOSEM**.
- Designed large-scale experimental pipelines for evaluating LLM behavior across multiple models, prompting strategies, datasets, and software engineering tasks, emphasizing reproducibility and statistically rigorous empirical evaluation.

INDUSTRY RESEARCH EXPERIENCE

Applied Scientist Intern Jun. 2026 – Sep. 2026
Amazon *Seattle, WA*

- Investigating machine learning approaches for large-scale workload forecasting, focusing on improving transaction-per-second (TPS) prediction accuracy.
- Developed data-driven business driver alignment methods to identify service-specific predictors of TPS, evaluating causal and statistical relationships among operational signals.
- Explored forecasting architectures and feature modeling strategies to improve prediction robustness beyond existing business-driver-based approaches.

Applied Scientist Intern Jun. 2025 – Sep. 2025
Amazon *Austin, TX*

- Investigated deployment risk assessment for large-scale software systems by analyzing code changes in Prime Video production repositories, developing machine learning approaches to identify potentially risky deployments.
- Designed a diff-aware feature framework that extracts code-level, change-level, and qualitative signals from software modifications to improve deployment risk prediction.
- Investigated the capability of LLMs for software change risk assessment by integrating diff-aware features into LLM-based classification pipelines and evaluating their effectiveness across production and open-source datasets.
- Demonstrated that carefully curated structural code complexity features provide stronger risk signals than traditional change-volume metrics, achieving an average F1-score of 0.81 across evaluated datasets.
- Drove the research project from idea development through experimentation and publication, resulting in acceptance at the **ASE 2026 Industry Showcase**.

Graduate Research Assistant Aug. 2023 – Dec. 2023
eBay *Remote*

- Investigated automated commit message generation using instruction-tuned open-source large language models for software engineering workflows.
- Evaluated prompt engineering and supervised fine-tuning strategies for repository-specific commit message generation across more than 50 software repositories.
- Improved commit message quality and stylistic consistency through instruction tuning and empirical evaluation of multiple open-source LLMs.

PUBLICATIONS

** indicates equal contribution.*

An Empirical Study on Evaluating Accessible Code Generation Capabilities of LLMs
Hyunjae Suh, Mahan Tafreshipour, Sam Malek, Iftekhar Ahmed
ACM Transactions on Software Engineering and Methodology (TOSEM), 2026. [DOI]

Deployment Risk Assessment Using Diff-Aware Features: A Case Study at Prime Video
Mayur Kurup*, **Hyunjae Suh***, Swathi Vaidyanathan, Pranesh Vyas, Srinidhi Madabhushi, Yegor Silyutin
IEEE/ACM International Conference on Automated Software Engineering (ASE), Industry Showcase, 2026. [arXiv]

An Empirical Study on Automatically Detecting AI-Generated Source Code: How Far Are We?
Hyunjae Suh, Mahan Tafreshipour, Jiawei Li, Adithya Bhattiprolu, Iftekhar Ahmed
IEEE/ACM International Conference on Software Engineering (ICSE), 2025. [DOI]

Does the Order of Fine-tuning Matter and Why?
Qihong Chen, Jiawei Li, **Hyunjae Suh**, Lianghao Jiang, Zheng Zhou, Jingze Chen, Jiri Gesi, Iftekhar Ahmed
arXiv preprint, 2024.

EDUCATION

University of California, Irvine
Ph.D. in Software Engineering — *Irvine, CA*

Kookmin University
B.S. in Computer Science — *Seoul, South Korea*

PROFESSIONAL SERVICE

Reviewer
ACM Transactions on Software Engineering and Methodology (TOSEM)
Empirical Software Engineering (EMSE)
IEEE Transactions on Reliability

TEACHING EXPERIENCE

Teaching Assistant — Sep. 2023 – Present
University of California, Irvine — *Irvine, CA*

- Supported instruction for undergraduate and graduate courses including Software Testing, Analysis, and Quality Assurance; Information Retrieval; Programming with Software Libraries; and How Computers Work.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, SQL

Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, Keras, Natural Language Processing (NLP)

Large Language Models: LLM Fine-tuning, Prompt Engineering, Agentic AI, ReAct, LLM Evaluation, RAG

Software Engineering Research: Program Analysis, Code Embeddings, Static Analysis, Software Metrics, Empirical Software Engineering

Cloud & Infrastructure: AWS (SageMaker, Bedrock, S3), Docker, Git, LaTeX

Software Engineering Tools: Soot, Understand, Checkstyle, ESLint, Detekt